

Components.css

```
/* =====  
COMPONENT LIBRARY  
===== */  
  
/* 1. TYPOGRAPHY & INPUTS */  
h1 { font-size: 2rem; font-weight: 800; margin-bottom: 10px; color: var(--navy); }  
h2.step-title { font-size: 1.2rem; text-transform: uppercase; color: #505a5f; border-  
bottom: 2px solid #b1b4b6; padding-bottom: 5px; margin-top: 30px; }  
p.instruction { font-size: 1.1rem; color: #505a5f; margin-bottom: 30px; }  
  
label { display: block; font-weight: 700; margin-bottom: 8px; font-size: 1rem; }  
  
input[type="text"], input[type="password"] {  
  width: 100%;  
  padding: 15px;  
  font-size: 1.2rem; /* LARGE TEXT */  
  border: 3px solid #0b0c0c;  
  border-radius: 0; /* Sharp edges = Authority */  
  box-sizing: border-box;  
  margin-bottom: 20px;  
}  
  
input:focus { outline: 4px solid var(--gold); border-color: var(--navy); }  
  
/* 2. BUTTONS */  
button {  
  width: 100%;  
  padding: 16px;  
  font-size: 1.1rem;  
  font-weight: 700;  
  border: none;  
  cursor: pointer;  
  text-transform: uppercase;  
  transition: transform 0.1s;  
}  
  
button:active { transform: scale(0.98); }  
  
.btn-primary { background-color: var(--navy); color: white; box-shadow: 0 4px 0  
#001122; }  
.btn-secondary { background-color: white; border: 3px solid var(--navy); color: var(--  
navy); margin-bottom: 15px; }  
.btn-accent { background-color: var(--gold); color: black; font-weight: 800; box-  
shadow: 0 4px 0 #b39700; }  
.btn-success { background-color: var(--green); color: white; margin-top: 10px; }  
.btn-outline { background: transparent; border: 2px solid #505a5f; color: #505a5f;  
margin-top: 10px; }
```

```
.btn-text { background: none; color: #505a5f; text-decoration: underline; margin-top: 40px; font-size: 0.9rem; }
```

```
/* 3. LOGIN SCREEN */
```

```
.login-panel { background: white; padding: 40px 20px; border: 1px solid #ddd; text-align: center; margin-top: 20px; }
```

```
.crest-placeholder {  
  width: 60px; height: 60px;  
  background: var(--navy); color: var(--gold);  
  border-radius: 50%; margin: 0 auto 20px;  
  display: grid; place-items: center; font-weight: 900;  
}
```

```
/* 4. TRAFFIC LIGHT CARDS (THE CORE VISUAL) */
```

```
.status-card {  
  padding: 25px;  
  border: 4px solid black;  
  margin-top: 20px;  
  color: white;  
}
```

```
.status-card h2 { font-size: 2.5rem; margin: 0; line-height: 1; }
```

```
.status-card p { font-size: 1.3rem; margin: 10px 0 0 0; font-weight: 600; }
```

```
.bg-green { background-color: var(--green); border-color: #004d29; }
```

```
.bg-orange { background-color: var(--orange); color: black; border-color: #b54914; }
```

```
.bg-red { background-color: var(--red); border-color: #8f1f0e; }
```

```
/* 5. AI BOX */
```

```
.ai-assistant-box {  
  background: #eef5ff;  
  border-left: 5px solid var(--navy);  
  padding: 15px;  
  margin-bottom: 20px;  
}
```

```
.ai-input-group { display: flex; gap: 10px; }
```

```
.ai-input-group input { margin-bottom: 0; }
```

```
.ai-input-group button { width: auto; padding: 0 20px; }
```

```
/* 6. TICKET PREVIEW */
```

```
.ticket-preview {  
  background: white;  
  border: 2px dashed var(--navy);  
  padding: 15px;  
  margin-top: 15px;  
}
```

```
.fine-text { display: block; font-size: 1.5rem; color: var(--red); font-weight: 800; text-align: right; }
```

Main.css

```
/* =====  
ZENTINEL THEME: AUTHORITATIVE & ACCESSIBLE  
"Function over Form"  
===== */
```

```
:root {  
  /* ZRP PALETTE */  
  --navy: #002244;  
  --gold: #FFD700;  
  --white: #FFFFFF;  
  --grey-bg: #F0F2F5;  
  
  /* SIGNAL COLORS */  
  --green: #00703C; /* Gov.uk Green */  
  --red: #D4351C; /* Gov.uk Red */  
  --orange: #F47738;  
  
  /* SPACING */  
  --pad: 20px;  
}
```

```
body {  
  margin: 0;  
  font-family: 'Inter', sans-serif;  
  background-color: var(--grey-bg);  
  color: #0b0c0c;  
  -webkit-font-smoothing: antialiased;  
}
```

```
/* HEADER: Always fixed at top context */  
header {  
  background: var(--navy);  
  color: var(--white);  
  padding: 15px var(--pad);  
}
```

```
.header-inner {  
  display: flex;  
  justify-content: space-between;  
  align-items: center;  
}
```

```
.brand { display: flex; flex-direction: column; }  
.zrp-text { font-size: 0.75rem; letter-spacing: 1px; opacity: 0.8; }  
.system-text { font-size: 1.2rem; font-weight: 800; letter-spacing: 0.5px; }
```

```
.gold-line {  
  height: 4px;
```

```
background: var(--gold);
margin-top: 10px;
}
```

```
#officer-badge {
font-size: 0.8rem;
font-weight: 600;
display: flex;
align-items: center;
gap: 8px;
}
```

```
.status-dot {
width: 10px; height: 10px;
background: var(--green);
border-radius: 50%;
box-shadow: 0 0 5px var(--green);
}
```

```
/* CONTAINER */
.app-container {
padding: var(--pad);
max-width: 600px; /* Tablet width limit for readability */
margin: 0 auto;
}
```

```
/* UTILS */
.hidden { display: none !important; }
```

```
/*
=====
=====
LOGIN STAGE 2: BIOMETRIC BUTTONS
=====
===== */
```

```
/* The Container for the buttons */
.bio-grid {
display: flex;
gap: 20px;
justify-content: center;
margin: 30px 0;
}
```

```
/* The Big Buttons (Face ID / Fingerprint) */
.btn-bio {
background: white;
border: 2px solid var(--zrp-gold);
```

```

border-radius: 15px;
width: 140px;
height: 140px;
display: flex;
flex-direction: column;
align-items: center;
justify-content: center;
cursor: pointer;
transition: transform 0.2s, box-shadow 0.2s;
color: var(--zrp-blue);
font-weight: 700;
box-shadow: 0 4px 10px rgba(0,0,0,0.1);
}

```

```

.btn-bio:hover {
transform: translateY(-5px);
background: #FFFAF0; /* Light cream hover */
box-shadow: 0 8px 20px rgba(0,0,0,0.15);
}

```

```

.btn-bio:active {
transform: scale(0.95);
}

```

```

.bio-icon {
font-size: 40px;
margin-bottom: 15px;
}

```

```

/* Success Banner (appears above buttons) */
.success-banner {
background: var(--green);
color: white;
padding: 10px;
border-radius: 8px;
margin-bottom: 20px;
font-weight: bold;
text-align: center;
animation: fadeIn 0.5s;
}

```

```

/* Ensure the 'hidden' class works */
.hidden {
display: none !important;
}

```

```

/*
=====
=====

```

BIOMETRIC ANIMATIONS (iPhone Style)

```
===== */
```

```
/* The Background Blur */
```

```
#bio-overlay {  
  position: fixed;  
  top: 0; left: 0; width: 100%; height: 100%;  
  background: rgba(0, 0, 0, 0.6);  
  backdrop-filter: blur(8px);  
  z-index: 9999;  
  display: flex;  
  justify-content: center;  
  align-items: center;  
  animation: fadeIn 0.3s ease-out;  
}
```

```
/* The Popup Box (Starts Blue) */
```

```
.bio-box {  
  width: 220px;  
  height: 220px;  
  background: rgba(30, 50, 90, 0.95); /* Deep Blue */  
  border-radius: 25px;  
  display: flex;  
  flex-direction: column;  
  align-items: center;  
  justify-content: center;  
  box-shadow: 0 20px 50px rgba(0,0,0,0.5);  
  transition: all 0.5s cubic-bezier(0.175, 0.885, 0.32, 1.275);  
  transform: scale(0.8);  
  animation: popIn 0.4s forwards;  
}
```

```
.bio-box.success { background: var(--green); transform: scale(1); }  
.bio-box.error { background: var(--red); animation: shake 0.4s; }
```

```
/* The Scanner Container */
```

```
.bio-scanner {  
  position: relative;  
  width: 80px;  
  height: 80px;  
  margin-bottom: 20px;  
}
```

```
/* 1. The Circulating Gold Ring */
```

```
.scan-ring {  
  position: absolute;  
  width: 100%; height: 100%;
```

```
border: 4px solid rgba(255, 215, 0, 0.2); /* Faint Gold */
border-top: 4px solid #FFD700; /* Bright Gold */
border-radius: 50%;
animation: spin 1s linear infinite;
}
```

```
/* 2. Face Features (Gold Lines) */
```

```
.face-features {
  position: absolute;
  top: 15px; left: 15px;
  width: 50px; height: 50px;
}

.eye {
  position: absolute; top: 15px; width: 6px; height: 6px;
  background: #FFD700; border-radius: 50%;
}

.eye.left { left: 10px; }
.eye.right { right: 10px; }
.mouth {
  position: absolute; bottom: 10px; left: 10px;
  width: 30px; height: 15px;
  border-bottom: 3px solid #FFD700;
  border-radius: 0 0 50% 50%;
}
```

```
/* 3. Fingerprint (CSS Only Representation) */
```

```
.fingerprint-lines {
  width: 50px; height: 60px;
  margin: 10px auto;
  border: 3px solid #FFD700;
  border-radius: 50% 50% 10px 10px;
  position: relative;
}

.fingerprint-lines::before {
  content: "; position: absolute;
  top: 5px; left: 5px; right: 5px; bottom: 5px;
  border: 3px solid rgba(255,215,0,0.5);
  border-radius: 50% 50% 5px 5px;
}
```

```
/* 4. Success Tick / Error Cross */
```

```
.result-icon {
  font-size: 60px;
  color: white;
  font-weight: bold;
  display: flex;
  justify-content: center;
  align-items: center;
  width: 100%; height: 100%;
}
```

```
text-shadow: 0 2px 10px rgba(0,0,0,0.2);
}
```

```
/* Text Styling */
#bio-text {
  color: white;
  font-size: 16px;
  font-weight: 600;
  margin: 0;
  text-align: center;
}
```

```
/* Keyframes */
@keyframes spin { 0% { transform: rotate(0deg); } 100% { transform: rotate(360deg); } }
@keyframes popIn { 0% { transform: scale(0.5); opacity: 0; } 100% { transform: scale(1); opacity: 1; } }
@keyframes shake { 0%, 100% { transform: translateX(0); } 25% { transform: translateX(-10px); } 75% { transform: translateX(10px); } }
@keyframes fadeIn { from { opacity: 0; } to { opacity: 1; } }
```

```
/* Helpers to hide/show specific icons */
.hidden { display: none !important; }
```

```
/*
=====
=====
COMBINED WORKSPACE STYLES
=====
===== */
```

```
.panel-section {
  background: white;
  padding: 15px;
  border-radius: 10px;
  margin-bottom: 20px;
  box-shadow: 0 2px 5px rgba(0,0,0,0.05);
  border: 1px solid #eee;
}
```

```
.checklist-grid {
  display: grid;
  grid-template-columns: 1fr 1fr; /* Two columns for checklist */
  gap: 10px;
}
```

```
.ai-section {
  border-left: 4px solid var(--zrp-gold);
```



```
}
```

```
.ai-input-wrapper textarea {  
  width: 100%;  
  padding: 12px;  
  border: 1px solid #ccc;  
  border-radius: 8px;  
  font-family: inherit;  
  font-size: 16px;  
  resize: vertical;  
}
```

```
#ticket-breakdown-list {  
  background: #f9f9f9;  
  padding: 15px;  
  border-radius: 8px;  
  margin-bottom: 20px;  
  border: 1px dashed #ccc;  
}
```

```
.breakdown-item {  
  display: flex;  
  justify-content: space-between;  
  padding: 8px 0;  
  border-bottom: 1px solid #eee;  
}
```

```
.breakdown-item:last-child {  
  border-bottom: none;  
}
```

```
.total-row {  
  display: flex;  
  justify-content: space-between;  
  font-size: 20px;  
  font-weight: 800;  
  color: var(--zrp-blue);  
  margin-bottom: 20px;  
  padding: 10px;  
  background: #eef2ff;  
  border-radius: 8px;  
}
```

Ticket.css

```
/* --- css/ticket.css --- */
```

```
/* The Container mimicking A4 Paper or the specific Slip */  
.ticket-wrapper {
```

```
background: #fdfbf7; /* Slight yellowing paper look */
color: #000;
font-family: "Courier New", Courier, monospace; /* Typewriter feel */
padding: 20px;
max-width: 800px;
margin: 20px auto;
border: 1px solid #ccc;
box-shadow: 0 10px 30px rgba(0,0,0,0.15);
position: relative;
overflow: hidden; /* For watermark */
}
```

```
/* Watermark Styles */
.watermark {
  position: absolute;
  top: 50%;
  left: 50%;
  transform: translate(-50%, -50%) rotate(-45deg);
  font-size: 8rem;
  font-weight: bold;
  color: rgba(255, 0, 0, 0.15); /* Faint Red */
  pointer-events: none;
  z-index: 0;
  white-space: nowrap;
  border: 10px solid rgba(255, 0, 0, 0.15);
  padding: 20px;
}
```

```
/* Header */
.ticket-header {
  text-align: center;
  border-bottom: 2px solid #333;
  padding-bottom: 10px;
  margin-bottom: 10px;
  position: relative;
}
```

```
.ticket-header h1 {
  font-size: 1.5rem;
  margin: 5px 0;
  text-transform: uppercase;
}
```

```
.ticket-header h3 {
  font-size: 1rem;
  margin: 0;
  font-weight: normal;
}
```

```
.ticket-number {  
  position: absolute;  
  top: 0;  
  right: 0;  
  font-size: 1.2rem;  
  font-weight: bold;  
  color: #d00; /* Red Serial Number */  
}
```

```
/* Table Structure */  
.ticket-table {  
  width: 100%;  
  border-collapse: collapse;  
  border: 2px solid #000;  
  position: relative;  
  z-index: 1; /* Above watermark */  
}
```

```
.ticket-table td {  
  border: 1px solid #000;  
  padding: 5px;  
  vertical-align: top;  
}
```

```
/* Field Labels vs Data */  
.field-box {  
  position: relative;  
  min-height: 40px;  
}
```

```
.label {  
  font-size: 0.7rem;  
  color: #444;  
  display: block;  
  margin-bottom: 2px;  
}
```

```
/* The "Blue Pen" Data */  
.fill-in, .fill-in-multi, .blue-ink {  
  font-family: "Courier Prime", "Courier New", monospace;  
  color: #00008b; /* Dark Blue */  
  font-size: 1rem; /* Adjusted for better fit */  
  font-weight: 700; /* Bold */  
  text-transform: uppercase;  
  letter-spacing: 0.05em; /* Spacing makes it very readable */  
}
```

```
.signature-font {  
  font-family: "Dancing Script", cursive;
```

```
font-size: 1.5rem;
}
```

```
/* Specific Layout Tweaks */
.description-box {
  height: 120px;
}
```

```
.date-time-row {
  margin-top: 20px;
  display: flex;
  justify-content: space-between;
  border-top: 1px dashed #999;
  padding-top: 5px;
}
```

```
.fine-box {
  background: #eee;
  text-align: right;
  font-size: 1.5rem;
}
```

```
.legal-text {
  font-size: 0.7rem;
  font-style: italic;
  padding: 10px;
  text-align: justify;
}
```

```
/* Action Buttons (Hide when printing) */
.final-actions {
  margin-top: 20px;
  display: flex;
  gap: 10px;
  justify-content: center;
}
```

```
@media print {
  .no-print { display: none; }
  body { background: white; }
  .ticket-wrapper { box-shadow: none; border: none; margin: 0; width: 100%; }
}
```

```
/* === STRICT PRINT STYLES === */
@media print {
  /* 1. Hide EVERYTHING by default */
  body * {
    visibility: hidden;
  }
}
```

```

/* 2. Hide specific UI containers just to be safe */
#view-patrol, #view-login, .final-actions, nav, header {
    display: none !important;
}

```

```

/* 3. Make the Ticket Visible */
#view-final-ticket,
#view-final-ticket .ticket-wrapper,
#view-final-ticket .ticket-wrapper * {
    visibility: visible;
    display: block;
}

```

```

/* 4. Position the Ticket at the very top-left of the paper */
.ticket-wrapper {
    position: absolute;
    left: 0;
    top: 0;
    width: 100%;
    margin: 0;
    padding: 0;
    border: none; /* Remove border for print if desired */
    box-shadow: none;
    background: white; /* Save ink */
}

```

```

/* 5. Ensure text colors print correctly */
.blue-ink {
    color: #00008b !important;
    -webkit-print-color-adjust: exact;
    print-color-adjust: exact;
}

.watermark {
    opacity: 0.1; /* Faint on paper */
}

```

ai_agent.js

```

/*
=====
=====
MODULE: LOCAL SEARCH AGENT (NO API REQUIRED)
File: js/modules/ai_agent.js
Description: Scans 'zrp_laws.txt' locally for keywords and has a built-in cheat sheet
for prototype-purposes only.

```

```
===== */  
  
let ZIMBABWE_LAW_TEXT = "";
```

```
// 1. LOAD THE LAWS (Background Process)
```

```
async function loadLaws() {  
  try {  
    const response = await fetch('zrp_laws.txt');  
    if (response.ok) {  
      ZIMBABWE_LAW_TEXT = await response.text();  
      console.log(`[ZRP Agent] Laws loaded successfully.`);  
    }  
  } catch (error) {  
    console.warn(`[ZRP Agent] Law file not found. Using internal cheat sheet  
only.`);  
  }  
}  
loadLaws();
```

```
export const AIAgent = {  
  analyze: async (userText) => {  
    console.log(`[ZRP Agent] Analyzing: "${userText}"`);
```

```
    // Fake "Thinking" delay for the UI (0.8 seconds)  
    await new Promise(r => setTimeout(r, 800));
```

```
    const query = userText.toLowerCase();
```

```
    // -----  
    // LEVEL 1: THE CHEAT SHEET (Guaranteed Hits)  
    // -----  
    // 1. SPEEDING  
    if (query.includes("speed") || query.includes("fast") || query.includes("limit")) {  
      return {  
        charge: "Exceeding Speed Limit",  
        act: "Road Traffic Act (Chapter 13:11)",  
        section: "Section 76",  
        fine_amount: 30, // Level 3 Fine  
        is_safety_critical: true,  
        description: "No person shall drive a vehicle upon any road at a speed in  
excess of the speed limit prescribed for that road."  
      };  
    }  
  }
```

```
    // 2. DRIVERS LICENSE
```

```
    if (query.includes("licence") || query.includes("license")) {  
      return {
```

```

        charge: "Driving Without a License",
        act: "Road Traffic Act (Chapter 13:11)",
        section: "Section 6",
        fine_amount: 20,
        is_safety_critical: false,
        description: "No person shall drive a motor vehicle on a road unless he is
the holder of a valid licence in respect of that class of motor vehicle."
    };
}

```

```

// 3. INDICATORS / SIGNALS
if (query.includes("signal") || query.includes("indicate") || query.includes("turn"))
{
    return {
        charge: "Failure to Signal",
        act: "Statutory Instrument 154",
        section: "Section 39",
        fine_amount: 20,
        is_safety_critical: true,
        description: "The driver of a vehicle on a road shall not turn or move
across the path of other traffic without giving a clear and sufficient signal."
    };
}
}

```

```

// 4. SEATBELT
if (query.includes("belt") || query.includes("strap")) {
    return {
        charge: "Failure to Wear Safety Belt",
        act: "Statutory Instrument 154",
        section: "Section 81",
        fine_amount: 10,
        is_safety_critical: true,
        description: "The driver and every passenger in a motor vehicle shall wear
a safety belt while the vehicle is in motion."
    };
}
}

```

```

// 5. DRUNK DRIVING
if (query.includes("drink") || query.includes("drunk") || query.includes("alcohol"))
{
    return {
        charge: "Driving Under Influence",
        act: "Road Traffic Act (Chapter 13:11)",
        section: "Section 55",
        fine_amount: 200, // Court Appearance
        is_safety_critical: true,
        description: "Any person who drives or attempts to drive a vehicle on a
road while under the influence of alcohol or drugs shall be guilty of an offence."
    };
}
}

```

```
}
```

```
// -----  
// LEVEL 2: THE FILE SCAN (For obscure stuff)  
// -----  
const fileMatch = findBestMatchInFile(query);  
if (fileMatch) {  
    return {  
        charge: "Traffic Offense (Detected)",  
        act: "Road Traffic Act (Chapter 13:11)",  
        section: "Refer to Description",  
        fine_amount: 15, // Default fine  
        is_safety_critical: false,  
        description: fileMatch  
    };  
}
```

```
// -----  
// LEVEL 3: THE FALLBACK (The "Catch-All")  
// -----  
return {  
    charge: "Driving Without Due Care",  
    act: "Road Traffic Act (Chapter 13:11)",  
    section: "Section 52",  
    fine_amount: 15,  
    is_safety_critical: true,  
    description: "Any person who drives a vehicle on a road without due care and  
attention or without reasonable consideration for other persons using the road shall  
be guilty of an offence."  
};  
}
```

```
// Internal Helper for scanning the file  
function findBestMatchInFile(query) {  
    if (!ZIMBABWE_LAW_TEXT) return null;  
  
    // Split file into paragraphs  
    const paragraphs = ZIMBABWE_LAW_TEXT.split(/\n\s*\n/);  
    const words = query.split(" ").filter(w => w.length > 3);  
  
    for (let para of paragraphs) {  
        let matches = 0;  
        for (let word of words) {  
            if (para.toLowerCase().includes(word)) matches++;  
        }  
        // If we find a paragraph with 2+ keywords, return it  
        if (matches >= 2) return para.trim().substring(0, 300) + "...";  
    }  
}
```



```
    return null;
  }
}
```

Auth.js

```
/*
=====
=====
MODULE: AUTHENTICATION (PROCESS 1.0)
File: js/modules/auth.js
Description: Handles User Login, RBAC, and Biometric Simulation.
=====
===== */
import { DB } from './db.js';

export const Auth = {
  // 1.1 Verify Credentials (Force ID & Password)
  login: async (forceId, password) => {
    console.log(`[Auth] Authenticating Force ID: ${forceId}`);

    // Simulating Network Delay
    await new Promise(r => setTimeout(r, 500));

    const user = DB.getUser(forceId);

    if (!user) {
      throw new Error("User not found.");
    }

    // 1.2 Verify Password (In production, use bcrypt comparison)
    if (user.password_hash !== password) {
      throw new Error("Invalid Password.");
    }

    console.log(`[Auth] Credentials Valid. Awaiting Biometrics...`);

    // Return user info so the UI knows who is trying to log in
    return {
      forceId: user.force_id,
      name: user.name,
      rank: user.rank,
      permissions: getPermissions(user.rank) // Prepare permissions
    };
  },

  // NEW: 1.4 Verify Biometrics (FaceID / Fingerprint)
  // This simulates the hardware scan delay and success signal
}
```

```

// NEW: 1.4 Verify Biometrics (With Simulation Logic)
verifyBiometric: async (scanType) => {
  console.log(`[Auth] Starting Biometric Scan: ${scanType.toUpperCase()}`);

  // 1. Simulate Scanning Delay
  await new Promise(r => setTimeout(r, 2000));

  // --- SIMULATION LOGIC ---
  // Math.random() returns a number between 0 and 1.
  // If it is less than 0.5, we pretend the scan failed.
  const isSuccess = Math.random() > 0.5;

  if (!isSuccess) {
    console.warn("[Auth] Biometric Scan Rejected.");
    throw new Error("Biometric Mismatch"); // This triggers the Red Box in app.js
  }

  // 2. Simulate Success
  return {
    success: true,
    method: scanType,
    token: `BIO-JWT-${Date.now()}-SECURE`,
    message: `${scanType === 'face' ? 'Face ID' : 'Fingerprint'} Verified`
  };
}

// RBAC Rules Definition
function getPermissions(rank) {
  if (rank === "Inspector") {
    return {
      canIssueTicket: true,
      canOverrideImpound: true, // Only Inspectors can release impounded cars
      viewAdminLogs: true
    };
  } else {
    // Constable
    return {
      canIssueTicket: true,
      canOverrideImpound: false, // Critical Restriction
      viewAdminLogs: false
    };
  }
}

```

Db.js

```

/*
=====

```

```

=====
MODULE: DATABASE (MOCK DATA LAYER)
File: js/modules/db.js
Description: Simulates the 4 Data Stores defined in DFD Level 1.
Updated: Expanded Statutory List for Robust MVP Demo.

=====
===== */

// --- 1. USERS DATA STORE (Task 2.1) ---
const USERS_STORE = [
  {
    force_id: "9921",
    password_hash: "password123",
    rank: "Constable",
    name: "Cst. M. Moyo",
    station_id: "HRE-central-01",
    biometric_token: "bio_fp_9921_valid"
  },
  {
    force_id: "1001",
    password_hash: "admin777",
    rank: "Inspector",
    name: "Insp. T. Chitepo",
    station_id: "HRE-hq-01",
    biometric_token: "bio_face_1001_valid"
  }
];

// --- 2. STATUTORY CODES STORE (Task 2.2 - ROBUST VERSION) ---
// Covers SI 154 (Equipment) and RTA Chapter 13:11 (Conduct)
const STATUTORY_STORE = [
  // === CATEGORY A: LICENSING & INSURANCE ===
  {
    code: "RTA-C13-S6",
    keywords: ["license", "licence", "dl", "driver"],
    act: "Road Traffic Act [Chapter 13:11]",
    section: "Section 6",
    charge: "Driving without a valid driver's license",
    fine_amount: 100.00,
    currency: "USD",
    is_safety_critical: true
  },
  {
    code: "RTA-C13-S22",
    keywords: ["insurance", "third party", "policy"],
    act: "Road Traffic Act [Chapter 13:11]",
    section: "Section 22",
    charge: "Driving without valid Third Party Insurance",
  }
];

```

```

    fine_amount: 30.00,
    currency: "USD",
    is_safety_critical: false
  },
  {
    code: "VR-ACT-S10",
    keywords: ["disc", "vehicle license", "zinara"],
    act: "Vehicle Registration & Licensing Act",
    section: "Section 10",
    charge: "Failure to display valid vehicle license disc",
    fine_amount: 20.00,
    currency: "USD",
    is_safety_critical: false
  },

```

```

// === CATEGORY B: TIRES & WHEELS (SI 154) ===
{
  code: "SI154-S14",
  keywords: ["tires", "tyres", "worn", "bald", "tread", "ply"],
  act: "Statutory Instrument 154 of 2010",
  section: "Section 14",
  charge: "Worn tires (tread depth less than 1mm)",
  fine_amount: 20.00, // Per tire
  currency: "USD",
  is_safety_critical: true
},
{
  code: "SI154-S16",
  keywords: ["spare", "wheel", "jack"],
  act: "Statutory Instrument 154 of 2010",
  section: "Section 16",
  charge: "No functional spare wheel or jack",
  fine_amount: 15.00,
  currency: "USD",
  is_safety_critical: false
},

```

```

// === CATEGORY C: LIGHTS & REFLECTORS ===
{
  code: "SI154-S18",
  keywords: ["headlight", "headlamp", "light", "dim"],
  act: "Statutory Instrument 154 of 2010",
  section: "Section 18",
  charge: "Operating vehicle without functional headlamps",
  fine_amount: 20.00,
  currency: "USD",
  is_safety_critical: true
},
{

```

```

    code: "SI154-S19",
    keywords: ["brake light", "stop light", "rear light"],
    act: "Statutory Instrument 154 of 2010",
    section: "Section 19",
    charge: "Non-functional stop/brake lights",
    fine_amount: 20.00,
    currency: "USD",
    is_safety_critical: true
  },
  {
    code: "SI154-S24",
    keywords: ["indicator", "turn signal", "flasher"],
    act: "Statutory Instrument 154 of 2010",
    section: "Section 24",
    charge: "Non-functional direction indicators",
    fine_amount: 15.00,
    currency: "USD",
    is_safety_critical: true
  },
  {
    code: "SI154-S35",
    keywords: ["reflector", "white", "front reflector"],
    act: "Statutory Instrument 154 of 2010",
    section: "Section 35",
    charge: "Missing White Front Reflectors",
    fine_amount: 10.00,
    currency: "USD",
    is_safety_critical: false
  },
  {
    code: "SI154-S36",
    keywords: ["honeycomb", "red", "rear reflector"],
    act: "Statutory Instrument 154 of 2010",
    section: "Section 36",
    charge: "Missing Red Rear Reflectors (Honeycomb)",
    fine_amount: 10.00,
    currency: "USD",
    is_safety_critical: false
  },

```

// === CATEGORY D: SAFETY EQUIPMENT ===

```

  {
    code: "SI154-S56",
    keywords: ["fire", "extinguisher", "fire extinguisher"],
    act: "Statutory Instrument 154 of 2010",
    section: "Section 56",
    charge: "No functional fire extinguisher (0.75kg min)",
    fine_amount: 20.00,
    currency: "USD",

```

```

    is_safety_critical: true
  },
  {
    code: "SI154-S57",
    keywords: ["triangle", "red triangle", "breakdown"],
    act: "Statutory Instrument 154 of 2010",
    section: "Section 57",
    charge: "Failure to carry two red reflective triangles",
    fine_amount: 15.00,
    currency: "USD",
    is_safety_critical: false
  },
  {
    code: "SI154-S61",
    keywords: ["vest", "reflective vest", "yellow vest"],
    act: "Statutory Instrument 154 of 2010",
    section: "Section 61",
    charge: "Failure to carry reflective safety vest",
    fine_amount: 10.00,
    currency: "USD",
    is_safety_critical: false
  },
],

```

// === CATEGORY E: DRIVING CONDUCT ===

```

{
  code: "RTA-C13-S51",
  keywords: ["speed", "speeding", "fast"],
  act: "Road Traffic Act [Chapter 13:11]",
  section: "Section 51",
  charge: "Exceeding speed limit",
  fine_amount: 50.00,
  currency: "USD",
  is_safety_critical: true
},
{
  code: "RTA-C13-S46",
  keywords: ["careless", "negligent", "lane"],
  act: "Road Traffic Act [Chapter 13:11]",
  section: "Section 46",
  charge: "Careless driving / Failure to maintain lane",
  fine_amount: 100.00, // Often court summons, but fine for MVP
  currency: "USD",
  is_safety_critical: true
},
{
  code: "RTA-C13-S72",
  keywords: ["obey", "police", "instruction", "stop"],
  act: "Road Traffic Act [Chapter 13:11]",
  section: "Section 72",

```

```
    charge: "Failure to obey police instruction",
    fine_amount: 300.00,
    currency: "USD",
    is_safety_critical: true
  }
];
```

```
// --- 3. VEHICLE HISTORY STORE (Task 2.3 - Internal) ---
```

```
let HISTORY_STORE = [
  {
    vrn: "XYZ-9999",
    violations: [
      { date: "2023-11-12", offense: "Worn Tires", status: "UNPAID" },
      { date: "2024-01-05", offense: "No Fire Extinguisher", status: "PAID" },
      { date: "2024-02-20", offense: "Broken Headlight", status: "UNPAID" }
    ]
  }
];
```

```
// --- 4. ZINARA REGISTRY (Task 2.3 - External Entity) ---
```

```
const ZINARA_EXTERNAL_DB = [
  {
    // SCENARIO 1: The "Perfect" Driver (Green Status)
    vrn: "ABC-1234",
    make: "Toyota Hilux",
    color: "White",
    owner: "Swift Logistics Ltd",
    license_expiry: "2026-12-31", // Future date
    insurance_status: "Valid",
    is_stolen: false,
    is_wanted: false
  },
  {
    // SCENARIO 2: The "Expired" Driver (Orange Status)
    vrn: "HRE-5555",
    make: "Honda Fit",
    color: "Silver",
    owner: "P. Ndlovu",
    license_expiry: "2023-01-01", // Past date
    insurance_status: "Expired",
    is_stolen: false,
    is_wanted: false
  },
  {
    // SCENARIO 3: The "Criminal" Vehicle (Red Status)
    vrn: "CRIME-001",
    make: "Nissan Navara",
    color: "Black",
    owner: "Unknown",
  }
];
```

```

    license_expiry: "2025-01-01",
    insurance_status: "Valid",
    is_stolen: true, // CRITICAL FLAG
    is_wanted: true // CRITICAL FLAG
  },
  {
    // SCENARIO 4: The "Habitual Offender" (Matches History Store)
    vrn: "XYZ-9999",
    make: "Mazda Demio",
    color: "Blue",
    owner: "T. Mambo",
    license_expiry: "2025-06-01",
    insurance_status: "Valid",
    is_stolen: false,
    is_wanted: false
  }
],

// =====
// PUBLIC METHODS (The Interface)
// =====
export const DB = {
  // 1. Fetch User (Sync)
  getUser: (forceld) => {
    return USERS_STORE.find(u => u.force_id === forceld);
  },

  // 2. Fetch ZINARA Record (Async Simulation)
  getZinaraRecord: (vrn) => {
    return new Promise((resolve, reject) => {
      setTimeout(() => {
        const record = ZINARA_EXTERNAL_DB.find(v => v.vrn === vrn);
        if (record) resolve(record);
        else reject("Vehicle not found in ZINARA Registry");
      }, 800);
    });
  },

  // 3. Fetch Internal History (Sync - Local Cache)
  getViolationHistory: (vrn) => {
    const entry = HISTORY_STORE.find(h => h.vrn === vrn);
    return entry ? entry.violations : [];
  },

  // 4. Find Law by Keyword (AI Lookup)
  findStatutoryCode: (description) => {
    const searchTerms = description.toLowerCase().split(" ");
    // Improved Matcher: Returns the law with the MOST matching keywords
    let bestMatch = null;

```



```

let maxMatches = 0;

STATUTORY_STORE.forEach(law => {
  const matches = law.keywords.filter(k =>
description.toLowerCase().includes(k)).length;
  if (matches > maxMatches) {
    maxMatches = matches;
    bestMatch = law;
  }
});
return bestMatch;
};

// Save Inspection Report to ZRP/ZINARA Cloud
saveInspectionResult: (report) => {
  console.log(`[Database] SYNCING REPORT TO ZRP CLOUD...`);
  console.log(`[Database] VRN: ${report.vrn}`);
  console.log(`[Database] Total Fines: ${report.totalFine}`);
  console.log(`[Database] Defect List:`, report.defects);

  // In a real app, this would be: await api.post('/inspections', report);
  return true;
}

```

generateticket.js

```

/* js/modules/generateTicket.js */

export function openFinalTicket(data) {
  // 1. Switch Views
  document.getElementById('view-patrol').classList.add('hidden');
  document.getElementById('view-final-ticket').classList.remove('hidden');

  // 2. Populate Fields
  setText('f265-station', data.officer.station || "HARARE CENTRAL");
  setText('f265-officer', `${data.officer.forcelid} ${data.officer.rank} ${
{data.officer.name}`);
  setText('f265-sig-officer', data.officer.name);

  // Offender
  const names = data.offender.name.split(' ');
  setText('f265-surname', names.pop() || data.offender.name);
  setText('f265-names', names.join(' ') || "----");
  setText('f265-address', data.offender.address);
  setText('f265-vrn', `${data.offender.vrn} / ${data.offender.make}`);

  // Offenses

```

```
const descBox = document.getElementById('f265-desc');
if (descBox) {
  descBox.innerHTML = data.offenses.map((o, i) =>
    `<div>${i+1}. ${o.desc.toUpperCase()}</div>`
  ).join("");
}
```

```
// Totals
setText('f265-total', data.total);
const now = new Date();
setText('f265-date', now.toLocaleDateString());
setText('f265-time', now.toLocaleTimeString());
```

```
window.scrollTo(0,0);
}
```

```
export function closeFinalTicket() {
  document.getElementById('view-final-ticket').classList.add('hidden');
  document.getElementById('view-patrol').classList.remove('hidden');
}
```

```
export function processFinalPayment(type) {
  // 1. Simple Alert (Mock Success)
  alert(`PAYMENT SUCCESSFUL\n\nMethod: ${type.toUpperCase()}\nStatus:
Processed\n\nReturning to Patrol...`);
```

```
// 2. Auto-Reset the App
// We trigger the click on the hidden reset button in app.js to handle the cleanup
const resetBtn = document.getElementById('btn-reset-all');
if (resetBtn) {
  resetBtn.click();
} else {
  // Fallback if button missing
  location.reload();
}
```

```
function setText(id, val) {
  const el = document.getElementById(id);
  if(el) el.textContent = val;
}
```

Inspection.js

```
import { DB } from './db.js';
```

```
/*
```

```
=====
=====
```

MODULE: INSPECTION (PROCESS 3.0)

Description: Manages SI 154 Compliance Checks and calculates fines.

```
=====
===== */
```

```
export const Inspection = {
```

```
  // 3.1 THE LAW LIBRARY (SI 154 CHECKLIST)
```

```
  // We added 'fine' (ZWG/USD) and 'code' to each item.
```

```
  getChecklist: () => {
```

```
    return [
```

```
      {
```

```
        id: "tires",
```

```
        label: "Tires (Tread < 1mm)",
```

```
        isCritical: true,
```

```
        fine: 30,
```

```
        code: "SI154-14(1)"
```

```
      },
```

```
      {
```

```
        id: "brakes",
```

```
        label: "Foot/Hand Brakes Defective",
```

```
        isCritical: true,
```

```
        fine: 60,
```

```
        code: "SI154-18"
```

```
      },
```

```
      {
```

```
        id: "lights",
```

```
        label: "Headlamps/Tail Lights Defective",
```

```
        isCritical: true,
```

```
        fine: 20,
```

```
        code: "SI154-22"
```

```
      },
```

```
      {
```

```
        id: "wipers",
```

```
        label: "No Windscreen Wipers",
```

```
        isCritical: false,
```

```
        fine: 10,
```

```
        code: "SI154-43"
```

```
      },
```

```
      {
```

```
        id: "horn",
```

```
        label: "No Warning Device (Horn)",
```

```
        isCritical: false,
```

```
        fine: 10,
```

```
        code: "SI154-46"
```

```
      },
```

```
      {
```

```
        id: "fire",
```

```

      label: "No Fire Extinguisher",
      isCritical: false,
      fine: 15,
      code: "SI154-52"
    },
    {
      id: "triangles",
      label: "Missing Warning Triangles",
      isCritical: false,
      fine: 15,
      code: "SI154-53"
    }
  ],
},

```

// 3.2 & 5.0 PROCESS & STORE RESULTS

```

submit: (vrn, failedItemIds) => {
  console.log(`[Inspection] Processing defects for ${vrn}...`);

```

// 1. Retrieve full details for the checked items

```

const allItems = Inspection.getChecklist();
const defects = allItems.filter(item => failedItemIds.includes(item.id));

```

// 2. Calculate Total Fine

// We use .reduce to sum up the 'fine' property of all defects

```

const totalFine = defects.reduce((sum, item) => sum + item.fine, 0);

```

// 3. Check for Impound Condition (Critical Failures)

```

const criticalDefects = defects.filter(item => item.isCritical);
const isImpound = criticalDefects.length > 0;

```

// 4. Create the Data Packet (The "Report")

```

const report = {
  id: `INS-${Date.now()}`,
  timestamp: new Date().toISOString(),
  vrn: vrn,
  status: isImpound ? "FAIL - CRITICAL" : (defects.length > 0 ? "FAIL - MINOR" : "PASS"),
  action: isImpound ? "IMPOUND" : (defects.length > 0 ? "TICKET" : "RELEASE"),
  defects: defects, // We store the whole array of failed items here
  totalFine: totalFine,
  notes: isImpound
    ? `Vehicle unsafe. ${criticalDefects.length} critical faults.`
    : "Routine inspection completed."
};

```

// 5. SAVE TO DATABASE (Persist the data)

```

DB.saveInspectionResult(report);

```

```

    // 6. Return to UI
    return report;
  }
};

```

Payment.js

```

/*
=====
=====
MODULE: PAYMENT HANDLER (PROCESS 6.0)
File: js/modules/payment.js
Description: Manages Financial Transactions and Offline/Online States.
=====
===== */

export const Payment = {

  // 6.1: Determine Available Payment Methods
  // Logic: If Offline, Digital Payments (EcoCash/Swipe) are REMOVED to prevent
  errors.
  getPaymentOptions: () => {
    const isOnline = navigator.onLine; // Browser API check

    console.log(`[Payment] Connectivity Status: ${isOnline ? 'ONLINE' : 'OFFLINE'}`);

    if (isOnline) {
      return [
        { id: 'ecocash', label: 'EcoCash (Mobile Money)', type: 'DIGITAL' },
        { id: 'innbucks', label: 'InnBucks', type: 'DIGITAL' },
        { id: 'swipe', label: 'ZIMSWITCH (Swipe)', type: 'DIGITAL' },
        { id: 'cash', label: 'Cash (Station Receipt)', type: 'MANUAL' },
        { id: 'form265', label: 'Issue Form 265 (Pay Later)', type: 'DEBT' }
      ];
    } else {
      // "Graceful Offline" Protocol:
      // Force the officer to use manual methods to avoid "Transaction Pending"
      limbo.

      console.warn("[Payment] System Offline. Digital Payments Disabled.");
      return [
        { id: 'cash', label: 'Cash (Offline Receipt)', type: 'MANUAL' },
        { id: 'form265', label: 'Issue Form 265 (Notice to Pay)', type: 'DEBT' }
      ];
    }
  },
};

```

```
// 6.2 & 6.4: Execute Transaction Logic
```

```
processPayment: async (ticketData, methodId, phoneNumber = null) => {  
  console.log(`[Payment] Initiating ${methodId} for ${ticketData.amount}...`);
```

```
  // SCENARIO A: DIGITAL PUSH (Online Only)
```

```
  if (methodId === 'ecocash' || methodId === 'innbucks') {
```

```
    if (!navigator.onLine) {
```

```
      throw new Error("Cannot process digital payment while offline.");
```

```
    }
```

```
    return simulateMobilePush(phoneNumber, ticketData.amount);
```

```
  }
```

```
  // SCENARIO B: DEBT NOTICE (Offline/Form 265)
```

```
  if (methodId === 'form265') {
```

```
    return generateForm265(ticketData);
```

```
  }
```

```
  // SCENARIO C: CASH (Manual)
```

```
  return {
```

```
    success: true,
```

```
    status: "PAID",
```

```
    receiptRef: `ZRP-CASH-${Date.now()}`,
```

```
    message: "Cash Payment Recorded. Receipt Generated.",
```

```
    method: "CASH"
```

```
  };
```

```
};
```

```
// -----
```

```
// HELPER: SIMULATE MOBILE MONEY PUSH (Process 6.2)
```

```
// -----
```

```
function simulateMobilePush(phone, amount) {
```

```
  return new Promise((resolve) => {
```

```
    console.log(`[Network] Sending USSD Push to ${phone}...`);
```

```
    
```

```
    // Simulate waiting for user to enter PIN on their phone
```

```
    setTimeout(() => {
```

```
      // 90% Success Rate Simulation
```

```
      const success = Math.random() > 0.1;
```

```
      
```

```
      if (success) {
```

```
        resolve({
```

```
          success: true,
```

```
          status: "PAID",
```

```
          receiptRef: `ECO-${Math.floor(Math.random() * 1000000)}`,
```

```
          message: "Transaction Approved. Funds Transferred to ZRP
```

```
Treasury.",
```

```
          method: "ECOCASH"
```

```
        });
```

```

    } else {
      resolve({
        success: false,
        status: "FAILED",
        message: "Transaction Timed Out or User Cancelled."
      });
    }
  }, 2000); // 2-second delay
});
}

// -----
// HELPER: GENERATE LIABILITY NOTICE (Process 6.4)
// -----
function generateForm265(ticketData) {
  const noticeNumber = `F265-${Date.now().toString().slice(-6)}`;

  // Logic: This creates a "Debt" record that syncs to ZINARA later
  // The "Form 265" is the legal document in Zimbabwe for "Notice to Pay Fine"
  console.log(`[Offline Protocol] Generated Liability Notice ${noticeNumber}`);

  return {
    success: true,
    status: "UNPAID", // Important: This flags the vehicle in the future
    receiptRef: noticeNumber,
    type: "DEBT_NOTICE",
    message: `Form 265 (${noticeNumber}) issued. Driver has 7 days to pay at any
ZRP Station.`,
    details: {
      dueDate: new Date(Date.now() + 7 * 24 * 60 * 60 * 1000).toDateString(),
      station: "Harare Central"
    }
  };
}

```

Vehicle.js

```

import { DB } from './db.js';

export const Vehicle = {
  // 2.0 Analyze Status
  analyze: async (vrn) => {
    console.log(`[Vehicle Analysis] Querying VRN: ${vrn}`);

    try {
      // 2.1 Fetch External Data (ZINARA)
      // Note: This fails if Offline, triggering the catch block
      const zinaraData = await DB.getZinaraRecord(vrn);
    }
  }
}

```

```
// 2.2 Fetch Internal History (ZRP Local Cache)
```

```
const history = DB.getViolationHistory(vrn);
```

```
// 2.3 Determine Status Color (The Decision Matrix)
```

```
const status = calculateStatus(zinaraData, history);
```

```
return {  
  found: true,  
  data: zinaraData,  
  history: history,  
  status: status  
};
```

```
  } catch (error) {  
    console.warn("[Vehicle Analysis] Offline or Not Found", error);  
    // Graceful Offline Fallback  
    return {  
      found: false,  
      status: {  
        headline: "VEHICLE NOT FOUND",  
        color: "GREY",  
        code: "OFFLINE OR NOT IN DATABASE",  
        message: "Proceed with Caution - Manual Check Recommended",  
      },  
    };  
  }  
}  
}  
};
```

```
// THE TRAFFIC LIGHT ALGORITHM
```

```
function calculateStatus(vehicle, history) {
```

```
  // PRIORITY 1: RED (CRITICAL DANGER)
```

```
  // Conditions: Stolen, Wanted, or Habitual Offender (>3 past violations)
```

```
  const isHabitualOffender = history.length >= 3;
```

```
  if (vehicle.is_stolen || vehicle.is_wanted || isHabitualOffender) {  
    return {  
      headline: "ALERT!",  
      color: "RED",  
      code: "HIGH RISK - IMPOUND REQUIRED",  
      message: vehicle.is_stolen ? "VEHICLE REPORTED STOLEN" : "HABITUAL  
OFFENDER - IMPOUND",  
      action: "ARREST_AND_IMPOUND"  
    };  
  }  
}
```

```
  // PRIORITY 2: ORANGE (ADMINISTRATIVE)
```

```
  // Conditions: Expired License, Invalid Insurance
```

```
  const isLicenseExpired = new Date(vehicle.license_expiry) < new Date();
```



```
const isInsuranceInvalid = vehicle.insurance_status !== "Valid";
```

```
if (isLicenseExpired || isInsuranceInvalid) {  
  return {  
    headline: "CAUTION",  
    color: "ORANGE",  
    code: "VIOLATION DETECTED",  
    message: isLicenseExpired ? "License Expired" : "Insurance Invalid",  
    action: "ISSUE_TICKET"  
  };  
}
```

```
// PRIORITY 3: GREEN (CLEAN)  
return {  
  headline: "VEHICLE CLEAR",  
  color: "GREEN",  
  code: "ALLOW TO PROCEED OR CONDUCT AN INSPECTION",  
  message: "Vehicle compliant and up to date with regulations.",  
  action: "PASS"  
};  
}
```

App.js

```
/*  
=====   
=====   
MAIN CONTROLLER (app.js)  
Status: COMPLETE FINAL  
Features: Split Workflows, Full Biometrics, Full Offline Support  
=====   
===== */
```

```
import { Auth } from './modules/auth.js';  
import { Vehicle } from './modules/vehicle.js';  
import { Inspection } from './modules/inspection.js';  
import { AIAgent } from './modules/ai_agent.js';  
import { Payment } from './modules/payment.js';  
import { openFinalTicket, closeFinalTicket, processFinalPayment } from './modules/  
generateTicket.js';
```

```
// --- APP STATE ---  
const STATE = {  
  currentUser: null,  
  currentVehicle: null,  
  currentTicket: null,  
  sessionOffenses: [], // The "Shopping Cart"  
  isOffline: !navigator.onLine
```

```
};
```

```
// --- DOM ELEMENTS ---
```

```
const UI = {  
  views: {  
    login: document.getElementById('view-login'),  
    patrol: document.getElementById('view-patrol'),  
    final: document.getElementById('view-final-ticket')  
  },  
  login: {  
    stage1: document.getElementById('login-stage-1'),  
    stage2: document.getElementById('login-stage-2'),  
    bioFeedback: document.getElementById('bio-feedback')  
  },  
  steps: {  
    workspace: document.getElementById('step-workspace'),  
    result: document.getElementById('step-result'),  
    actions: document.getElementById('step-actions')  
  },  
  inputs: {  
    forceId: document.getElementById('force-id'),  
    password: document.getElementById('password'),  
    vrn: document.getElementById('vrn-input')  
  },  
  header: {  
    badge: document.getElementById('officer-badge'),  
    name: document.getElementById('officer-name'),  
    dot: document.querySelector('.status-dot')  
  }  
};
```

```
/*
```

```
=====
```

1. INITIALIZATION & LISTENERS

```
=====
```

```
===== */  
document.addEventListener('DOMContentLoaded', () => {
```

```
// --- LOGIN FLOW ---
```

```
const loginForm = document.getElementById('login-form');  
if (loginForm) loginForm.addEventListener('submit', handleCredentialCheck);  
  
const btnFace = document.getElementById('btn-face-scan');  
if (btnFace) btnFace.addEventListener('click', () => handleBiometric('face'));  
  
const btnFinger = document.getElementById('btn-finger-scan');  
if (btnFinger) btnFinger.addEventListener('click', () =>
```

```

handleBiometric('fingerprint'));

const btnImpound = document.getElementById('btn-confirm-impound');
if (btnImpound) btnImpound.addEventListener('click', () => {
    alert("Impound Logged. Vehicle Status Updated.");
    resetPatrol();
});
// --- PATROL FLOW ---
document.getElementById('btn-search').addEventListener('click', () => {
    resetSessionData();
    handleSearch();
});

document.getElementById('vrn-input').addEventListener('input', function(e) {
    e.target.value = e.target.value.toUpperCase();
});

// --- WORKSPACE NAVIGATION ---
document.getElementById('btn-inspect').addEventListener('click',
openInspectionView);
document.getElementById('btn-ticket').addEventListener('click', openTicketView);

// --- ADDING OFFENSES ---
document.getElementById('btn-add-inspection').addEventListener('click',
addInspectionToCart);
document.getElementById('btn-add-traffic').addEventListener('click',
addTrafficToCart);

// --- NEW NAVIGATION LISTENERS ---
document.getElementById('btn-switch-to-ticket').addEventListener('click',
openTicketView);
document.getElementById('btn-switch-to-inspect').addEventListener('click',
openInspectionView);

// --- FINAL GENERATION ---
document.getElementById('btn-generate-ticket').addEventListener('click',
showSummaryBox);

// --- PROCEED TO FINAL FORM ---
const btnProceed = document.getElementById('btn-proceed-to-final');
if (btnProceed) btnProceed.addEventListener('click', handleProceedToFinal);

// --- FINAL FORM ACTIONS (FIXED) ---
const btnPayOnline = document.getElementById('btn-final-pay-online');
if (btnPayOnline) btnPayOnline.addEventListener('click', () =>
processFinalPayment('online'));

const btnPrint = document.getElementById('btn-final-print');
// FIX: Directly call window.print() for the browser print dialog

```

```
if (btnPrint) btnPrint.addEventListener('click', () => window.print());
```

```
const btnBackSummary = document.getElementById('btn-back-to-summary');  
if (btnBackSummary) btnBackSummary.addEventListener('click',  
closeFinalTicket);
```

```
const btnFinish = document.getElementById('btn-finish-patrol');  
if (btnFinish) btnFinish.addEventListener('click', resetPatrol);
```

```
// --- UTILITIES ---  
const btnReset = document.getElementById('btn-reset-all') ||  
document.getElementById('btn-reset');  
if (btnReset) btnReset.addEventListener('click', resetPatrol);
```

```
// --- SYSTEM UTILS ---  
window.addEventListener('online', updateNetworkStatus);  
window.addEventListener('offline', updateNetworkStatus);  
updateNetworkStatus();  
});
```

```
/*  
=====
```

2. LOGIN & BIOMETRICS

```
===== */
```

```
async function handleCredentialCheck(e) {  
  e.preventDefault();  
  const forceld = UI.inputs.forceld.value;  
  const password = UI.inputs.password.value;  
  const submitBtn = document.querySelector('#login-form button');
```

```
  try {  
    submitBtn.textContent = "VERIFYING...";  
    const userPartial = await Auth.login(forceld, password);  
    STATE.currentUser = userPartial;
```

```
    UI.login.stage1.classList.add('hidden');  
    UI.login.stage2.classList.remove('hidden');  
  } catch (error) {  
    alert("ACCESS DENIED: " + error.message);  
    submitBtn.textContent = "VERIFY CREDENTIALS";  
  }  
}
```

```
async function handleBiometric(type) {  
  const overlay = document.getElementById('bio-overlay');
```

```
const box = document.getElementById('bio-box');
const text = document.getElementById('bio-text');
const facelcon = document.querySelector('.face-features');
const fingerlcon = document.querySelector('.fingerprint-lines');
const resultlcon = document.querySelector('.result-icon');
const ring = document.querySelector('.scan-ring');
```

```
overlay.classList.remove('hidden');
box.className = 'bio-box';
resultlcon.classList.add('hidden');
ring.classList.remove('hidden');
text.textContent = type === 'face' ? "Scanning Face..." : "Scanning Print...";
```

```
if (type === 'face') {
  facelcon.classList.remove('hidden');
  fingerlcon.classList.add('hidden');
} else {
  facelcon.classList.add('hidden');
  fingerlcon.classList.remove('hidden');
}
```

```
try {
  const result = await Auth.verifyBiometric(type);
```

```
  if (result.success) {
    box.classList.add('success');
    ring.classList.add('hidden');
    facelcon.classList.add('hidden');
    fingerlcon.classList.add('hidden');
    resultlcon.textContent = "✓";
    resultlcon.classList.remove('hidden');
    text.textContent = "Verified!";
```

```
    setTimeout(() => {
      overlay.classList.add('hidden');
      completeLogin();
    }, 1500);
  }
} catch (error) {
  box.classList.add('error');
  ring.classList.add('hidden');
  facelcon.classList.add('hidden');
  fingerlcon.classList.add('hidden');
  resultlcon.textContent = "✗";
  resultlcon.classList.remove('hidden');
  text.textContent = "Unrecognised!";
```

```
    setTimeout(() => {
      overlay.classList.add('hidden');
```

```

    }, 2000);
  }
}

```

```

function completeLogin() {
  const user = STATE.currentUser;
  UI.header.name.textContent = `${user.rank} ${user.name}`;
  UI.header.badge.classList.remove('hidden');
  document.getElementById('view-login').classList.add('hidden');
  document.getElementById('view-login').classList.remove('active-view');
  document.getElementById('view-patrol').classList.remove('hidden');
  document.getElementById('view-patrol').classList.add('active-view');
}

```

```

/*
=====
=====
3. VEHICLE SEARCH
=====
===== */

```

```

async function handleSearch() {
  // 1. GET & CLEAN INPUT
  const inputEl = document.getElementById('vrn-input');
  let rawInput = inputEl.value.toUpperCase().trim();

  // Remove existing dashes/spaces/special chars to get "CRIME001"
  const cleanStr = rawInput.replace(/[^A-Z0-9]/g, "");

```

```

  if (cleanStr.length === 0) return alert("Please enter a VRN");

```

```

  // 2. SMART FORMATTING (Flexible)
  // Regex explanation:
  // ^([A-Z]+) -> Capture any number of letters at the start (Group $1)
  // ([0-9]+)$ -> Capture any number of digits at the end (Group $2)
  let formattedVrn = cleanStr.replace(/^[A-Z]+([0-9]+)$/, '$1-$2');

```

```

  // 3. UPDATE UI
  inputEl.value = formattedVrn;

```

```

  // 4. RESET UI STATE
  document.getElementById('step-result').classList.remove('hidden');
  document.getElementById('step-result').innerHTML = '<p>Scanning Database...</p>';
  document.getElementById('step-actions').classList.add('hidden');
  document.getElementById('step-impound').classList.add('hidden');

```

// 5. PERFORM ANALYSIS

```
const analysis = await Vehicle.analyze(formattedVrn);
STATE.currentVehicle = analysis;
```

// 6. RENDER CARD

```
const bgClass = analysis.status.color === 'RED' ? 'bg-red' : analysis.status.color
=== 'ORANGE' ? 'bg-orange' : 'bg-green';
document.getElementById('step-result').innerHTML = `
  <div class="status-card ${bgClass}">
    <h2>${analysis.status.headline}</h2>
    <p>${analysis.status.message}</p>
  </div>`;
```

// 7. HANDLE RED / IMPOUND LOGIC

```
if (analysis.status.color === 'RED') {
  document.getElementById('step-actions').classList.add('hidden');
  document.getElementById('step-impound').classList.remove('hidden');

  if (!STATE.currentUser.permissions?.canOverrideImpound) {
    // alert("⚠️ HIGH RISK VEHICLE DETECTED - IMPOUND REQUIRED");
  }
} else {
  document.getElementById('step-impound').classList.add('hidden');
  document.getElementById('step-actions').classList.remove('hidden');
}
}
```

```
/*
=====
=====
4. WORKFLOW: ADDING ITEMS
=====
===== */
```

```
function openInspectionView() {
  UI.steps.workspace.classList.remove('hidden');
  document.getElementById('workspace-inspection').classList.remove('hidden');
  document.getElementById('workspace-ticket').classList.add('hidden');
  document.getElementById('ticket-confirm-box').classList.add('hidden');

  const container = document.getElementById('checklist-container');
  if (container.children.length === 0) {
    container.innerHTML = '';
    Inspection.getChecklist().forEach(item => {
      const div = document.createElement('div');
      div.innerHTML = `
        <label style="display:flex; align-items:center; gap:10px; padding:5px;">
          <input type="checkbox" value="${item.id}" data-fine="${item.fine}" data-
```

```

label="${item.label}">
    <span>${item.label} (${item.fine})</span>
</label>`;
container.appendChild(div);
});
}

UI.steps.workspace.scrollToView({ behavior: 'smooth' });
}

```

```

function openTicketView() {
    UI.steps.workspace.classList.remove('hidden');
    document.getElementById('workspace-inspection').classList.add('hidden');
    document.getElementById('workspace-ticket').classList.remove('hidden');
    document.getElementById('ticket-confirm-box').classList.add('hidden');

    UI.steps.workspace.scrollToView({ behavior: 'smooth' });
    setTimeout(() => {
        const input = document.getElementById('traffic-ai-input');
        if(input) input.focus();
    }, 500);
}

```

```

async function addInspectionToCart() {
    const btn = document.getElementById('btn-add-inspection');
    btn.textContent = "Adding...";
    btn.disabled = true;

    const checkboxes = document.querySelectorAll('#checklist-container
input[type="checkbox"]:checked');
    let newItems = [];

```

```

checkboxboxes.forEach(cb => {
    newItems.push({
        type: "Defect (Manual)",
        desc: cb.getAttribute('data-label'),
        fine: parseInt(cb.getAttribute('data-fine'))
    });
    cb.checked = false;
});
}

```

```

const textInput = document.getElementById('inspect-ai-input').value;
if (textInput.trim() !== "") {
    const phrases = textInput.split(/ and |,/i);
    const aiPromises = phrases.map(async (phrase) => {
        if (phrase.trim().length < 3) return null;
        return await AIAgent.analyze(phrase.trim());
    });
}

```



```

    const results = await Promise.all(aiPromises);
    results.forEach(res => {
        if (res) newItems.push({ type: "Defect (AI)", desc: res.charge, fine:
res.fine_amount });
    });
}

```

```

addToSession(newItems);
document.getElementById('inspect-ai-input').value = "";
btn.textContent = "➕ ADD FINDINGS TO TICKET";
btn.disabled = false;
}

```

```

async function addTrafficToCart() {
    const btn = document.getElementById('btn-add-traffic');
    const input = document.getElementById('traffic-ai-input');
    if(!input.value) return alert("Please describe the offense first.");

```

```

    btn.textContent = "Analyzing...";
    btn.disabled = true;

```

```

    try {
        const result = await AIAgent.analyze(input.value);
        const offense = [{
            type: "Traffic Violation",
            desc: `${result.charge} (${result.act})`,
            fine: result.fine_amount
        }];
    }

```

```

        addToSession(offense);
        input.value = "";

```

```

    } catch (e) {
        alert("AI could not identify offense. Please try rephrasing.");
    }

```

```

    btn.textContent = "➕ ADD OFFENSE TO TICKET";
    btn.disabled = false;
}

```

```

function addToSession(newItems) {
    if (newItems.length === 0) return;

```

```

    const uniqueItems = newItems.filter(newItem => {
        const exists = STATE.sessionOffenses.some(existing =>
            existing.desc === newItem.desc
        );
    });

```

```
    return !exists;
  });
```

```
  if (uniqueItems.length === 0) {
    return alert("Item already exists on the ticket list.");
  }
```

```
  STATE.sessionOffenses = [...STATE.sessionOffenses, ...uniqueItems];
  document.getElementById('action-bar-final').classList.remove('hidden');
  if (uniqueItems.length < newItem.length) {
    alert(`Added ${uniqueItems.length} new item(s). Duplicates were excluded.`);
  } else {
    alert(`Added ${uniqueItems.length} item(s) to ticket compilation.`);
  }
}
```

```
/*
=====
=====
5. FINAL STAGE: GENERATE SUMMARY & PROCEED
=====
===== */
```

```
function showSummaryBox() {
  document.getElementById('workspace-inspection').classList.add('hidden');
  document.getElementById('workspace-ticket').classList.add('hidden');
  document.getElementById('action-bar-final').classList.add('hidden');

  const listContainer = document.getElementById('ticket-breakdown-list');
  listContainer.innerHTML = "";

  STATE.sessionOffenses.forEach((item, index) => {
    const div = document.createElement('div');
    div.className = 'summary-item';
    div.style.display = 'flex';
    div.style.justifyContent = 'space-between';
    div.style.borderBottom = '1px dashed #ccc';
    div.style.padding = '5px 0';

    div.innerHTML = `
      <span>${index+1}. ${item.desc}</span>
      <strong>$$${item.fine}</strong>
    `;
    listContainer.appendChild(div);
  });
}
```

```
const total = STATE.sessionOffenses.reduce((sum, item) => sum + item.fine, 0);
document.getElementById('lbl-total-fine').textContent = `$$${total}.00`;
```

```
const finalBox = document.getElementById('ticket-confirm-box');
finalBox.classList.remove('hidden');
finalBox.scrollToView({ behavior: 'smooth' });
```

```
}
```

```
function handleProceedToFinal() {
  const ticketData = {
    officer: {
      name: STATE.currentUser ? STATE.currentUser.name : "OFFICER",
      rank: STATE.currentUser ? STATE.currentUser.rank : "CST",
      forceId: STATE.currentUser ? STATE.currentUser.forceId : "9921",
      station: "HRE CENTRAL TRAFFIC"
    },
    offender: {
      name: STATE.currentVehicle ? STATE.currentVehicle.data.owner :
"UNKNOWN DRIVER",
      address: "1 Mock Address Ave, Harare, Zimbabwe",
      vrn: UI.inputs.vrn.value || "Unknown",
      make: STATE.currentVehicle ? STATE.currentVehicle.data.make : "N/A"
    },
    offenses: STATE.sessionOffenses,
    total: STATE.sessionOffenses.reduce((sum, item) => sum + item.fine, 0),
    dateTime: new Date()
  };
}
```

```
openFinalTicket(ticketData);
```

```
}
```

```
/*
=====
=====
6. HELPERS
=====
===== */
```

```
function resetSessionData() {
  STATE.sessionOffenses = [];
  document.getElementById('action-bar-final').classList.add('hidden');
  document.getElementById('ticket-breakdown-list').innerHTML = "";
  document.getElementById('lbl-total-fine').textContent = '$0.00';
  document.getElementById('ticket-confirm-box').classList.add('hidden');
}
```

```
function renderStatusCard(status) {
  const bgClass = status.color === 'RED' ? 'bg-red' :
```

```

        status.color === 'ORANGE' ? 'bg-orange' : 'bg-green';
    UI.steps.result.innerHTML = `
        <div class="status-card ${bgClass}">
            <h2>${status.headline || status.color}</h2>
            <p>${status.message}</p>
            <small style="display:block; margin-top:10px; opacity:0.8">${status.code}</
small>
        </div>
    `;
    UI.steps.result.classList.remove('hidden');
}

```

```

function resetPatrol() {
    STATE.currentVehicle = null;
    STATE.currentTicket = null;
    resetSessionData();
}

```

```

    UI.inputs.vrn.value = "";
    document.getElementById('checklist-container').innerHTML = "";

    const inspectInput = document.getElementById('inspect-ai-input');
    if(inspectInput) inspectInput.value = "";

    const trafficInput = document.getElementById('traffic-ai-input');
    if(trafficInput) trafficInput.value = "";

    // Hide everything
    const successMsg = document.getElementById('payment-success-msg');
    if(successMsg) successMsg.classList.add('hidden');

```

```

    document.getElementById('step-result').classList.add('hidden');
    document.getElementById('step-actions').classList.add('hidden');
    document.getElementById('step-impound').classList.add('hidden'); // <--- ADD
THIS LINE
    document.getElementById('step-workspace').classList.add('hidden');
    document.getElementById('ticket-confirm-box').classList.add('hidden');

    if (UI.views.final) UI.views.final.classList.add('hidden');
    UI.views.patrol.classList.remove('hidden');

```

```

    window.scrollTo({ top: 0, behavior: 'smooth' });
}

```

```

function updateNetworkStatus() {
    STATE.isOffline = !navigator.onLine;
    if (STATE.isOffline) {
        UI.header.dot.style.background = 'grey';
        UI.header.name.textContent += ' (OFFLINE)';
    } else {

```

```

    Ul.header.dot.style.background = 'var(--green)';
    if (STATE.currentUser) {
      Ul.header.name.textContent = `${STATE.currentUser.rank} $
{STATE.currentUser.name}`;
    }
  }
}
}

```

Index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Zentinel | ZRP Digital Enforcement</title>

  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link href="https://fonts.googleapis.com/css2?
family=Inter:wght@400;600;800&family=Caveat:wght@700&family=Courier+Prime:
wght@400;700&display=swap" rel="stylesheet">

  <link rel="stylesheet" href="css/main.css">
  <link rel="stylesheet" href="css/ticket.css">
  <link rel="stylesheet" href="css/components.css">
</head>
<body>

```

```

  <header id="app-header">
    <div class="header-inner">
      <div class="brand">
        <span class="zrp-text">ZIMBABWE REPUBLIC POLICE</span>
        <span class="system-text">ZENTINEL</span>
      </div>
      <div id="officer-badge" class="hidden">
        <span class="status-dot"></span>
        <span id="officer-name">OFFLINE</span>
      </div>
    </div>
    <div class="gold-line"></div>
  </header>

```

```

  <main class="app-container">
    <section id="view-login" class="active-view">
      <div class="login-panel">
        <div class="crest-placeholder">ZRP</div>
        <h2 style="color:var(--zrp-blue); margin-bottom: 5px;">OFFICER LOGIN</
h2>

```

`<p style="color:#666; font-size:12px; margin-bottom:20px;">Restricted
Access: ZRP Traffic Patrol Officers Only.</p>`

```
<div id="login-stage-1">
  <form id="login-form">
    <label>FORCE NUMBER</label>
    <input type="text" id="force-id" placeholder="e.g. 9921" required>
    <label>PASSWORD</label>
    <input type="password" id="password" placeholder="....."
required>
    <button type="submit" class="btn-primary" style="margin-top:15px;">
      VERIFY CREDENTIALS
    </button>
  </form>
</div>
```

```
<div id="login-stage-2" class="hidden" style="text-align: center;">
  <p style="color: green; font-weight: bold; margin-bottom: 15px;">
Credentials Verified</p>
  <p style="font-size: 14px; margin-bottom: 20px;">Complete biometric
authentication to proceed.</p>
  <button id="btn-face-scan" class="btn-secondary" style="width:100%;
margin-bottom:10px;">
    <span></span> SCAN FACE
  </button>
  <button id="btn-finger-scan" class="btn-secondary"
style="width:100%;">
    <span></span> SCAN FINGERPRINT
  </button>
</div>
</div>
</section>
```

```
<section id="view-patrol" class="hidden">
  <div class="step-container" id="step-identify">
    <h2 class="step-title">1. Identify Vehicle</h2>
    <div class="search-group">
      <input type="text" id="vrn-input" placeholder="ENTER VRN (e.g.
ABC-1234)" autocapitalize="characters">
      <button id="btn-search" class="btn-primary">SEARCH DATABASE</
button>
    </div>
  </div>
  <div id="step-result" class="hidden"></div>
```

```

<div id="step-actions" class="hidden action-panel">
  <h2 class="step-title">2. Select Procedure</h2>
  <div class="grid-buttons">
    <button id="btn-inspect" class="btn-secondary">
      <span class="icon"></span> RUN INSPECTION ON VEHICLE
    </button>
    <button id="btn-ticket" class="btn-secondary">
      <span class="icon"></span> ISSUE TICKET FOR TRAFFIC
  </div>
</div>

```

OFFENSE

```

<div id="step-workspace" class="hidden workspace-panel">
  <div id="workspace-inspection" class="hidden">
    <h2 class="step-title">3. Digital Inspection (SI 154)</h2>
    <div class="panel-section">
      <h4>Standard Checklist</h4>
      <div id="checklist-container" class="checklist-grid"></div>
    </div>
    <div class="panel-section ai-section">
      <h4>Other</h4>
      <div class="ai-input-wrapper">
        <textarea id="inspect-ai-input" rows="2" placeholder="Describe
defects (e.g. Broken windshield)..."></textarea>
      </div>
    </div>
    <button id="btn-add-inspection" class="btn-primary">
      + ADD FINDINGS TO TICKET
    </button>
    <div style="margin-top: 15px; border-top: 1px solid #ccc; padding-top:
10px; text-align: center;">
      <p style="font-size: 0.8rem; color: #666;">Need to add a traffic
offense?</p>
      <button id="btn-switch-to-ticket" style="background: none; border:
1px solid var(--zrp-blue); color: var(--zrp-blue); padding: 8px 20px; border-radius:
4px; font-weight: bold; cursor: pointer;">
        SWITCH TO TRAFFIC OFFENSE
      </button>
    </div>
    <div id="workspace-ticket" class="hidden">
      <h2 class="step-title">3. Traffic Offense</h2>
      <div class="panel-section ai-section" style="border-left-color: var(--

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zrp-blue);">
    <h4>Describe Violation</h4>
    <div class="ai-input-wrapper">
        <textarea id="traffic-ai-input" rows="3" placeholder="Describe
offense (e.g. Speeding in school zone)..."></textarea>
    </div>
</div>

    <button id="btn-add-traffic" class="btn-secondary">
        + ADD OFFENSE TO TICKET
    </button>
    <div style="margin-top: 15px; border-top: 1px solid #ccc; padding-top:
10px; text-align: center;">
        <p style="font-size: 0.8rem; color: #666;">Need to inspect vehicle
defects?</p>
        <button id="btn-switch-to-inspect" style="background: none;
border: 1px solid var(--zrp-blue); color: var(--zrp-blue); padding: 8px 20px; border-
radius: 4px; font-weight: bold; cursor: pointer;">
            SWITCH TO INSPECTION
        </button>
    </div>
</div>

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    <div id="action-bar-final" class="hidden" style="margin-top: 20px; text-
align: center;">
        <p style="color: green; margin-bottom: 10px;">✓ Items added.</p>
        <button id="btn-generate-ticket" class="btn-success full-width">
            GENERATE OFFENSE SUMMARY
        </button>
    </div>
</div>

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    <div id="ticket-confirm-box" class="hidden">
        <div class="divider"></div>
        <h2 style="text-align: center; color: var(--zrp-blue);">OFFENSE
SUMMARY</h2>
        <div id="ticket-breakdown-list"></div>

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    <div class="total-row">
        <span>TOTAL PAYABLE:</span>
        <span id="lbl-total-fine">$0.00</span>
    </div>

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    <button id="btn-proceed-to-final" class="btn-primary full-width"
style="margin-top: 15px; font-weight: 800;">
        PROCEED TO FINAL TICKET →
    </button>
    <button id="btn-reset-all" class="btn-text" style="margin-top: 10px;

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width:100%;">Cancel / Reset</button>
</div>
</section>
```

```
<section id="view-final-ticket" class="hidden">
  <div class="ticket-wrapper">
    <div class="watermark">DEMO UNOFFICIAL</div>
    <div class="ticket-header">
      <div class="zrp-logo"></div>
      <div class="header-text">
        <h3>ZIMBABWE REPUBLIC POLICE</h3>
        <h1>ADMISSION OF GUILT TRAFFIC FINE</h1>
        <small>COURT FORMS: Z69J</small>
      </div>
      <div class="ticket-number">
        <span class="red-text">1264615 A</span>
      </div>
    </div>
```

```
<table class="ticket-table">
  <tr>
    <td colspan="2" class="field-box">
      <span class="label">Station:</span>
      <div id="f265-station" class="fill-in">---</div>
    </td>
    <td colspan="2" class="field-box">
      <span class="label">Arresting Detail (Name/Rank/No):</span>
      <div id="f265-officer" class="fill-in">---</div>
    </td>
  </tr>
  <tr>
    <td colspan="4" class="field-box">
      <span class="label">Surname of person admitting guilt:</span>
      <div id="f265-surname" class="fill-in">---</div>
      <span class="label" style="margin-left:20px;">Other names:</
span>
      <div id="f265-names" class="fill-in">---</div>
    </td>
  </tr>
  <tr>
    <td colspan="3" class="field-box">
      <span class="label">Residential Address:</span>
      <div id="f265-address" class="fill-in">---</div>
    </td>
    <td class="field-box">
      <span class="label">VRN / Make:</span>
      <div id="f265-vrn" class="fill-in">---</div>
    </td>
  </tr>
```

```

        </td>
      </tr>
      <tr>
        <td colspan="4" class="field-box">
          <span class="label">Charge (section and statutory enactment):</span>
span>
          <div class="fill-in">SEC 4(1) OF THE R.T.A</div>
        </td>
      </tr>
      <tr>
        <td colspan="4" class="field-box description-box">
          <span class="label">Brief description of offence(s):</span>
          <div id="f265-desc" class="fill-in-multi"></div>
          <div class="date-time-row">
            <span>Date: <strong id="f265-date" class="blue-ink">---</strong></span>
strong></span>
            <span>Time: <strong id="f265-time" class="blue-ink">---</strong></span>
strong></span>
            <span>Place: <strong class="blue-ink">HARARE</strong></span>
span>
          </div>
        </td>
      </tr>
      <tr>
        <td colspan="3" class="field-box right-align">
          <span class="label">AMOUNT OF DEPOSIT FINE FIXED BY
PRESCRIBED OFFICER:</span>
        </td>
        <td class="field-box fine-box">
          <span class="currency">$</span>
          <span id="f265-total" class="amount blue-ink">0.00</span>
        </td>
      </tr>
      <tr>
        <td colspan="4" class="legal-text">
          I, the above named, do hereby admit that I am guilty of the said
offence(s) and wish to deposit the sum of money fixed.
        </td>
      </tr>
      <tr>
        <td colspan="2" class="field-box">
          <span class="label">Witnessed by (Officer):</span>
          <div id="f265-sig-officer" class="fill-in signature-font">---</div>
        </td>
        <td colspan="2" class="field-box">
          <span class="label">Signed (Offender):</span>
          <div class="fill-in signature-font">Digitally Signed via Biometric</div>
div>

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        </td>
      </tr>
    </table>
```

```
      <div class="final-actions no-print">
        <button id="btn-final-pay-online" class="btn-primary"> PAY NOW
(ONLINE)</button>
        <button id="btn-final-print" class="btn-secondary">PRINT TICKET</
button>
        <button id="btn-back-to-summary" class="btn-text">Modify / Go Back</
button>
      </div>
    </div>
  </section>
```

```
</main>
```

```
<div id="payment-success-msg" class="hidden">
  <div class="success-content">
    <div class="big-check">✓</div>
    <h2>PAYMENT SUCCESSFUL</h2>
    <p>Receipt #992211 sent to driver.</p>
    <button id="btn-finish-patrol" class="btn-primary">START NEW SEARCH</
button>
  </div>
</div>

<div id="bio-overlay" class="hidden">
  <div class="bio-box" id="bio-box">
    <div class="bio-scanner">
      <div class="scan-ring"></div>
      <div class="scan-icon face-features">
        <div class="eye left"></div>
        <div class="eye right"></div>
        <div class="mouth"></div>
      </div>
      <div class="scan-icon fingerprint-lines hidden"></div>
      <div class="scan-icon result-icon hidden"></div>
    </div>
    <h3 id="bio-text">Authenticating...</h3>
  </div>
</div>

<div id="step-actions" class="hidden action-panel">
  <h2 class="step-title">2. Select Procedure</h2>
  <div class="grid-buttons">
    <button id="btn-inspect" class="btn-secondary">
      <span class="icon"></span> RUN INSPECTION ON VEHICLE
    </button>
    <button id="btn-ticket" class="btn-secondary">
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```
        <span class="icon"></span> ISSUE TICKET FOR TRAFFIC OFFENSE
    </button>
</div>
</div>
```

```
<div id="step-impound" class="hidden action-panel" style="border: 2px solid
#d32f2f; background: #ffebee; padding: 20px; text-align: center; border-radius: 8px;
margin-top: 20px;">
    <h2 style="color: #c62828; font-weight: 800;">IMPOUND REQUIRED</h2>
    <p style="color: #333; margin-bottom: 20px;">
        This vehicle is flagged as <strong>HIGH RISK</strong>. <br>
        Do not issue a standard ticket. Secure the vehicle immediately.
    </p>
    <button id="btn-confirm-impound" class="btn-primary" style="background-color:
#c62828; width: 100%;">
        CONFIRM IMPOUND COMPLETE
    </button>
</div>
```

```
<script type="module" src="js/app.js"></script>
</body>
</html>
```